KCC CODE CLUB · GAME MAKERS

Week 2

Events and Movement

Week 2 of 8

Name _____

Today's big idea

Code doesn't only run when you click the green flag. It runs when **things happen** — a key press, a click, anything. Those triggers are called **events**.

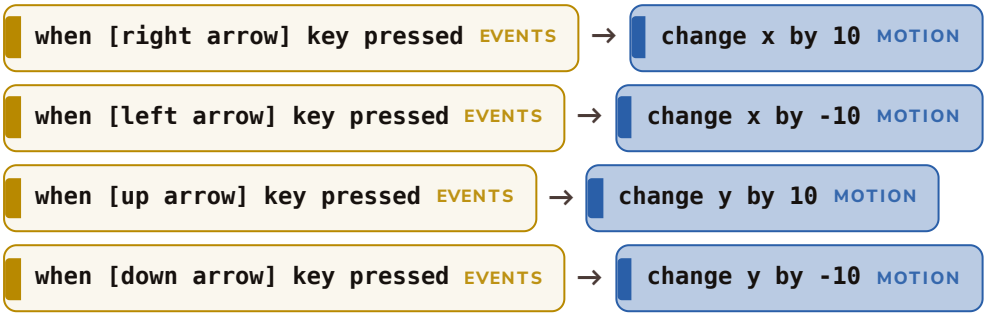
And position is just two numbers: **x** goes across, **y** goes up and down. The middle of the stage is `x: 0 y: 0`.

Words to know

Word	What it means
event	Something happening that makes code run
hat block	The curved block at the top of a script. It says *when* to run
coordinates	The two numbers that say where something is
change vs set	`change` adds to what's there. `set` replaces it

Class activity — Drive the Car

Make a sprite you can drive with all four arrow keys.



Now change the `10` to `50`. Too fast? Try `2`. Too slow? **Find the number that feels right.** That's called tuning, and it's a real part of making games.

Your project — the player moves

- [ ] Arrow keys move your player left and right
- [ ] The player stays near the bottom of the stage
- [ ] `go to x: 0 y: -140` under the green flag so it always starts in the same place
- [ ] Tune the speed until it feels good

- [ ] **Save it**

Homework

1. Add another dimension. Either let your player move up and down too, or make it wrap around — go off one side, come back on the other.

2. Face the right way. Use `point in direction` so your player faces where it's going. *(If your sprite ends up upside down, look at the rotation style in the sprite panel — the fix isn't in the code!)*

Bonus challenges

1. Add WASD keys as well as the arrows, so two people could play at once.
2. **Treasure Hunt.** Three or four sprites, one hiding treasure. The empty ones say `Nothing here!` when clicked. The winner says `You found it!` and grows by 50.
3. Add the **Pen** extension. Make your sprite draw a trail. Add a key that clears it.
4. Make the sprite move faster while you hold shift. *(Tricky! Come back to this in two weeks.)*

If something breaks

- **Moves once then stops?** You used `set x to` instead of `change x by`.
- **Nothing happens?** Wrong sprite selected when you wrote the code.
- **Gone off the screen?** Add `go to x: 0 y: -140` and click the green flag.

Stuck? Put your hand up — that is what we are here for. And always save before you close the lid.